

Pointers

Find out the output for the following programs.

```
1) #include <stdio.h >
   int main()
   {
       int *p = 10;
       printf(" %u\n", (unsigned int)p);
       printf("%d\n",*p);
   }
```

```
2) #include <stdio.h>
   int main()
   {
       int *ptr, a = 10;
       ptr = &a;
       *ptr += 1;
       printf("%d,%d/n", *ptr, a);

   }
```

```
3) #include<stdio.h>
   int main()
   {
       int x = -300;
       unsigned char *p;
       p = &x;
       printf("%d\n",*p++);
       printf("%d\n",*p);
   }
```

```
4) #include<stdio.h>
   int main()
   {
       int x = 256;
       char *p = &x;
       *++p = 2;
       printf("%d",x);
   }
```

```
5) #include<stdio.h>
   int main()
   {
       int x = 300;
       if(*(char *)&x == 44)
           printf("Little Endian\n");
       else
           printf("Big Endian\n");
   }
```

```
6) #include <stdio.h>
void main()
{
    int x = 0;
    int *ptr = &5;
    printf("%p\n", ptr);
}
```

```
7) #include<stdio.h>
int main()
{
    int const *p = 5;
    int q;
    p = &q;
    printf("%d",++(*p));
}
```

```
8) #include<stdio.h>
int main()
{
    int x = 10;
    int const * const p;
    p = &x;
    printf("%d\n", *p);
}
```

```
9) #include <stdio.h>
int x = 0;
void main()
{
    int *const ptr = &x;
    printf("%p\n", ptr);
    ptr++;
    printf("%p\n ", ptr);
}
```

```
10) #include <stdio.h>
int main()
{
    const int ary[4] = {1, 2, 3, 4};
    int *p;
    p = ary + 3;
    *p = 5;
    printf("%d\n", ary[3]);
}
```

```
11) #include <stdio.h>
int main()
{
    int ary[4] = {1, 2, 3, 4};
    int *p = ary + 3;
```

```
printf("%d\n", p[-2]); }
```

```
12) #include <stdio.h>
void main()
{
    char *s= "hello";
    char *p = s + 2;
    printf("%c\t%c", *p, s[1]);
}
```

```
13) #include <stdio.h>
int main()
{
    void *p;
    int a[4] = {1, 2, 3, 4};
    p = &a[3];
    int *ptr = &a[2];
    int n = (int*)p - ptr;
    printf("%d\n", n);
}
```

```
14) #include<stdio.h>
int main()
{
    int a[ ] = {10,20,30,40,50},i;
    char *p = a;

    for(i=0;i<5;i++)
    printf("%d ",*p++);
}
```

```
15) #include<stdio.h>
int main()
{
    int a[]={10,20,30,40,50};
    char *p;
    p=(char *)a;
    printf("%d\n",*((int *)p+4));
}
```

```
16) #include <stdio.h>
int main()
{
    double *ptr = (double *)100;
    ptr = ptr + 2;
    printf("%u\n", ptr);
}
```

```
17) #include <stdio.h>
int main()
{
```

```

    int i = 10;
    void *p = &i;
    printf("%d\n", (int *)*p);
    // printf("%d\n", *(int*)p);
    return 0;
}

```

18) #include <stdio.h>

```

int main()
{
    int a[4] = {1, 2, 3, 4};
    void *p = &a[1];
    void *ptr = &a[2];
    int n = 1;
    n = ptr - p;
    printf("%d\n", n);
}

```

19) #include <stdio.h>

```

int main()
{
    int *p = (int *)2;
    int *q = (int *)3;
    printf("%d", p + q);
}

```

20) Which of the following operand can be applied to pointers p and q?
 (Assuming initialization as int *a = (int *)2; int *b = (int *)3;)

- a) a + b
- b) a - b
- c) a * b
- d) a / b

Ans: b)

21) Which of following logical operation can be applied to pointers?
 (Assuming initialization int *a = 2; int *b = 3;)

- a) a | b
- b) a ^ b
- c) a & b
- d) None of the mentioned

Ans: d)

22) #include <stdio.h>

```

void main()
{
    char *s = "hello";
    char *n = "cjn";
    char *p = s + n;
    4 printf("%c\t%c", *p, s[1]);
}

```

```

23) #include <stdio.h>
void m(int *p)
{
    int i = 0;
    for(i = 0; i < 5; i++)
        printf("%d\t", p[i]);
}
void main()
{
    int a[5] = {6, 5, 3};
    m(&a);
}

```

```

24) #include <stdio.h>
void foo(int*);
int main()
{
    int i = 10, j = 20, *p = &i;
    foo(p++);
    foo(p);
}
void foo(int *p)
{
    printf("%d\n", *p);
}

```

```

25) #include <stdio.h>
int main()
{
    int i = 97, *p = &i;
    foo(&i);
    printf("%d ", *p);
}
void foo(int *p)
{
    int j = 2;
    p = &j;
    printf("%d ", *p);
}

```

```

26) #include <stdio.h>
int main()
{
    const int ary[4] = {1, 2, 3, 4};
    int *p = ary + 3;
    *p = 5;
    ary[3] = 6;
    printf("%d", ary[3]);
}

```

```

27) #include <stdio.h>

```

```

int main()
{
    char *p = "Hai friends", *p1 = p;
    while(*p != '\0');
    ++*p++;
    printf("%s %s\n", p, p1);
}

```

28) #include<stdio.h>

```

int main()
{
    char *x = "VECTOR";
    printf("%s\n", x+3);
    printf("%d\n"+1, 123456);
}

```

29) #include<stdio.h>

```

int main()
{
    char a[ ] = "abcdefgh";
    int *ptr = a;
    printf("%x %x\n", ptr[0], ptr[1]);
}

```

30) #include<stdio.h>
#include<string.h>

```

int main()
{
    char *str = "hello, world\n";
    char *strc = "good morning\n";
    strcpy(strc, str);
    printf("%s\n", strc);
    return 0;
}

```

31) #include <stdio.h>

```

int main()
{
    char *str = "hello world";
    char strc[50] = "good morning india\n";
    strcpy(strc, str);
    printf("%s\n", strc);
    return 0;
}

```

32) #include <stdio.h>

```

int main()
{
    char *str = "hello, world\n";
    str[5] = '.';
}

```

```

    printf("%s\n", str);
    return 0;
}

```

```

33) #include <stdio.h>
int main()
{
    char str[] = "hello, world";
    str[5] = '.';
    printf("%s\n", str);
    return 0;
}

```

```

34) #include <stdio.h>
int main()
{
    char *str = "hello world";
    char strary[] = "hello world";
    printf("%d %d\n", sizeof(str), sizeof(strary));
    return 0;
}

```

```

35) #include <stdio.h>
int main()
{
    char *str = "hello world";
    char strary[] = "hello world";
    printf("%d %d\n", strlen(str), strlen(strary));
    return 0;
}

```

```

36) #include<stdio.h>
int main()
{
    int a = 5,b = 4,c = 9;
    *(a>b ? &a : &b) = (a+b)>c;
    printf("%d %d\n",a,b);
}

```

37) Find the sizeof any datatype with out using sizeof operator. (Hint : Use pointers)

```

38) #include<stdio.h>
int main()
{
    int i;
    double a = 5.2;
    char *ptr;
    ptr = (char *)&a;
    for(i=0;i<=7;i++)
    printf("%d\n",*ptr++);
    return 0;
}

```

39) Correct the following program.

```
#include<stdio.h>
int main()
{
    void *p;
    int **ptr;
    int a = 129;
    p = &a;
    ptr = &p;
    printf(" p = %d  p = %u  &p = %u\n", *p, p, &p);
}
```

40) #include<stdio.h>

```
main()
{
    char a[20];
    char *p,*q;
    p=&a[0];
    q=&a[10];
    printf("%d %d\n",q-p,&q-&p);
}
```

41) #include<stdio.h>

```
main()
{
    int a=0x12345678;
    void *ptr;
    ptr=&a;
    printf("0x%x\n",*(int *)&*(&(char*)ptr));
}
```

42) #include<stdio.h>

```
main()
{
    int a[5]={1,2,3,4,5};
    int *ptr=(int *)(&a+1);
    printf("%d %d\n",*(a+1),*(ptr-1));
    printf("%d %d\n",*(a+1),*(ptr));
}
```

43) #include <stdio.h>

```
void main()
{
    char *s= "hello";
    char *p = s;
    printf("%c\t%c", 1[p], s[1]);
}
```

44) #include<stdio.h>

```
main()
```

```

{
    char a[]="abcde";
    char *p=a;
    p++;
    p++;
    p[2]='z';
    printf("%s",p);
}

```

```

45) #include<stdio.h>
    main()
    {
        char a[]="ABCDEFGHIJKLMNOPQRSTUVWXYZ";
        int i,*p = a;
        for(i=0;i<5;i++)
            printf("%d\t",*p++);
    }

```

```

46) #include<stdio.h>
    main()
    {
        char a[]="abcdef";
        char *ptr1 = a;
        ptr1 = ptr1+(strlen(ptr1)-1);
        printf("%c", --*ptr1--);
        printf("%c",--*--ptr1);
        printf("%c",--*(ptr1--));
        printf("%c",--*(-ptr1));
        printf("%c",*ptr1);
    }

```

```

47) #include<stdio.h>
    int main()
    {
        char *str1 = "Hello";
        char *str2 = "Hai";
        char *str3;
        str3 = strcat(str1,str2);
        printf("%s %s\n",str3,str1);
        return 0;
    }

```

```

48) #include<stdio.h>
    int main()
    {
        char a[]="Hello";
        char *p="Hai";
        a="Hai";
        p="Hello";
        printf("%s %s\n",a,p);
        return 0;
    }

```

- 49) `#include<stdio.h>`
`int main()`
`{`
`int i,n;`
`char *x="Alice";`
`n=strlen(x);`
`*x=x[n];`
`for(i=0;i<=n;i++)`
`{`
`printf("%s",x);`
`x++;`
`}`
`printf("%s\n",x);`
`return 0;`
`}`
- 50) `#include<stdio.h>`
`char *str="char *str=%c%s%c;main(){printf(str,34,str,34);}";`
`int main()`
`{`
`printf(str,34,str,34);`
`return 0;`
`}`
- 51) `#include <stdio.h>`
`void f(char *k)`
`{`
`k++;`
`k[2] = 'm';`
`printf("%c\n", *k);`
`}`
`void main()`
`{`
`char s[] = "hello";`
`f(s);`
`printf("%s\n",s);`
`}`
- 52) `#include<stdio.h>`
`void t1(char *q);`
`main()`
`{`
`char *p;`
`p = "abcder";`
`t1(p);`
`}`
`void t1(char *q)`
`{`
`if(*q!='r')`
`{`

```

        putchar(*q);
        t1(q++);
    }
}

```

```

53) #include<stdio.h>
    int main(){
        int i;
        float a=5.2;
        char *ptr;
        ptr=(char *)&a;
        for(i=0;i<=3;i++)
            printf("%d ",*ptr++);
        return 0;
    }

```

```

54) #include <stdio.h>
    void foo( int[] );
    int main()
    {
        int ary[4] = {1, 2, 3, 4};
        foo(ary);
        printf("%d ", ary[0]);
    }
    void foo(int p[4])
    {
        int i = 10;
        p = &i;
        printf("%d ", p[0]);
    }

```

```

55) #include <stdio.h>
    void main()
    {
        int k = 5;
        int *p = &k;
        int **m = &p;
        **m = 10;
        printf("%d%d%d\n", k, *p, **m);
    }

```

```

56) #include <stdio.h>
    int main()
    {
        int a = 1, b = 2, c = 3;
        int *ptr1 = &a, *ptr2 = &b, *ptr3 = &c;
        int **sptr = &ptr1;
        printf("%d ",**sptr);
        *sptr = ptr2;
        printf("%d ",**sptr);
    }

```

```
57) #include <stdio.h>
void main()
{
    int a[3] = {1, 2, 3};
    int *p = a;
    int *r = &p;
    printf("%d\n", (**r));
}
```

```
58) #include <stdio.h>
int main()
{
    int i = 97, *p = &i;
    foo(&p);
    printf("%d ", *p);
    return 0;
}
void foo(int **p)
{
    int j = 2;
    *p = &j;
    printf("%d ", **p);
}
```

```
59) #include <stdio.h>
void foo(int *const *p);
int main()
{
    int i = 11;
    int *p = &i;
    foo(&p);
    printf("%d ", *p);
}
void foo(int *const *p)
{
    int j = 10;
    *p = &j;
    printf("%d ", **p);
}
```

```
60) #include <stdio.h>
void foo(int **const p);
int main()
{
    int i = 10;
    int *p = &i;
    foo(&p);
    printf("%d ", *p);
}
void foo(int **const p)
```

```

{
    int j = 11;
    *p = &j;
    printf("%d ", **p);
}

```

```

61) #include <stdio.h>
int *f();
int main()
{
    int *p = f();
    printf("%d\n", *p);
}
int *f()
{
    int *j = (int*)malloc(sizeof(int));
    *j = 10;
    return j;
}

```

```

62) #include <stdio.h>
void main()
{
    char *a[10] = {"hi", "hello", "how"};
    int i = 0;
    for (i = 0; i < 10; i++)
        printf("%s ", *(a[i]));
}

```

```

63) #include <stdio.h>
void main()
{
    char *a[10] = {"hi", "hello", "how"};
    int i = 0, j = 0;
    a[0] = "hey";
    for (i = 0; i < 10; i++)
        printf("%s ", a[i]);
}

```

```

64) #include <stdio.h>
void main()
{
    char *a[10] = {"hi", "hello", "how"};
    printf("%d\n", sizeof(a));
}

```

```

65) #include <stdio.h>
void main()
{
    char *a[10] = {"hi", "hello", "how"};
    printf("%d\n", sizeof(a[1]));
}

```

- 66) #include <stdio.h>
int main()
{
char a[2][6] = {"hello", "hi"};
printf("%s ", *a + 1);
return 0;
}
- 67) #include <stdio.h>
int main()
{
char *a[2] = {"hello", "hi"};
printf("%s\n", *(a + 1));
return 0;
}
- 68) #include <stdio.h>
int main(int argc, char *argv[])
{
while (argc--)
printf("%s\n", argv[argc]);
return 0;
}
- 69) #include <stdio.h>
int main(int argc, char *argv[])
{
while (*argv++ != NULL)
printf("%s\n", *argv);
return 0;
}
- 70) #include <stdio.h>
int main(int argc, char *argv[])
{
while (*argv != NULL)
printf("%s\n", *(argv++));
return 0;
}
- 71) #include<stdio.h>
int main(int sizeofargv, char *argv[])
{
while(sizeofargv)
printf("%s ",argv[--sizeofargv]);
return 0;
} if i/p is sample friday tuesday sunday
- 72) #include<stdio.h>

```

int main()
{
    char *str[]={“Progs”,”Do”,”Not”,”Die”,”They”,”Croak!”};
    printf(“%d %d”,sizeof(str),strlen(str[0]));
    return 0;
}

```

73) `#include<stdio.h>`

```

int main()
{
    static char *s[]={“black”,”white”,”pink”,”violet”};
    char **ptr[]={s+3,s+2,s+1,s},***p;
    p = ptr;
    printf(“%s\n”,**p+1);
    return 0;
}

```

74) `#include<stdio.h>`

```

main()
{
    char *m[]={“jan”,”feb”,”mar”};
    char d[][10] = {“sun”,”mon”,”tue”};
    printf(“%s\t”,m[1]);
    printf(“%s\t”,d[1]);
}

```

75) `#include<stdio.h>`

```

void fun(char **);
int main()
{
    char *argv[]={“ab”,”cd”,”ef”,”gh”};
    fun(argv);
    return 0;
}
void fun(char **p)
{
    char *t;
    t=(p+=sizeof(int))[-1];
    printf(“%s\n”,t);
}

```

76) `#include <stdio.h>`

```

void first()
{
    printf("first");
}
void second()
{
    first();
}
void third()
{

```

```

    second();
}
void main()
{
    void (*ptr)();
    ptr = third;
    ptr();
}

```

```

77) #include <stdio.h>
int add(int a, int b)
{
    return a + b;
}
int main()
{
    int (*fn_ptr)(int, int);
    fn_ptr = add;
    printf("The sum of two numbers is: %d\n", (int)fn_ptr(2, 3));
}

```

```

78) #include <stdio.h>
int mul(int a, int b, int c)
{
    return a * b * c;
}
void main()
{
    int (*function_pointer)(int, int, int);
    function_pointer = mul;
    printf("The product of three numbers is:%d",
    function_pointer(2, 3, 4));
}

```

```

79) #include<stdio.h>
int fun(int (*)());
int main()
{
    fun(main);
    printf("Hi\n");
    return 0;
}
int fun(int (*p)())
{
    printf("Hello\n");
    return 0;
}

```

```

80) #include<stdio.h>
int main()
{
    char *p = "Hello World";
}

```

```
printf(p);
```

```
}
```

```
-----END-----
```